Project A Final Report

‘Achtung, die Kurve!’

Tiny Tapeout ASIC Project

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Background & Motivation

Overcoming stringent silicon area limits to build a fully functional, multiplayer hardware game.

The Core Challenge

Application Specific Integrated Circuits (ASICs) represent the pinnacle of digital design, but open-source platforms like Tiny Tapeout impose strict standard-cell boundaries.

- Storing a 640x480 high-resolution video framebuffer within internal flip-flops is physically impossible in a 1x1 or 1x2 tile constraint.
- 'Achtung, die Kurve!' requires storing persistent trails, meaning the entire board state must be constantly read and rendered.
- **Solution:** We developed a custom QSPI PSRAM controller (APS6404L) to offload video memory to an external chip.

Why External PSRAM Is Essential

A full VGA framebuffer is small for memory - but enormous when built from standard-cell flip-flops.

FULL-RESOLUTION MONOCHROME

1 bit / pixel

640 × 480 = 307,200 pixels

307,200 DFFs **37.5 KiB**

≈571 TT tiles

at TT's standard 60% placement target

343 tiles at an area-only 100% packing lower bound

FULL-RESOLUTION, FOUR-STATE FRAMEBUFFER

2 bits / pixel

307,200 pixels × 2 bits

614,400 DFFs **75 KiB**

≈1,142 TT tiles

at TT's standard 60% placement target

686 tiles at an area-only 100% packing lower bound

75 KiB OFF-CHIP → THE FULL GAME FITS IN 1×2

Only the QSPI controller and game logic consume silicon; the framebuffer lives in external memory.

Optimistic area-only estimate: 20.019 μm^2/DFF, 17,955 μm^2/tile, 60% density. Routing, clocking, and control logic would require more.

What is Tiny Tapeout?

An educational project democratizing access to silicon manufacturing by lowering costs and simplifying the toolchain.

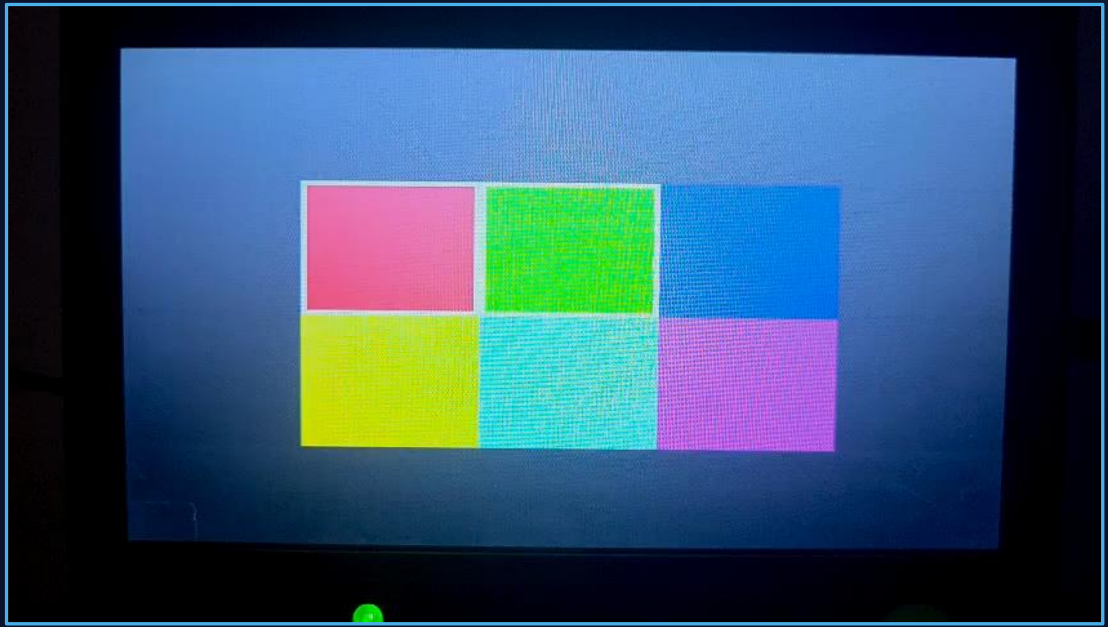
- **Multi-Project Wafer (MPW): Combines multiple independent user designs onto a single physical silicon chip to share manufacturing costs.**
- **Open-Source Tools:** Relies entirely on open-source Electronic Design Automation (EDA) flows, such as LibreLane and the sky130 PDK.
- **Area Constraints:** Designs are strictly limited to predefined "tiles" (e.g., 1x1 or 1x2 footprints), severely restricting standard cell capacity and routing resources.
- **Educational Focus:** Empowers students, hobbyists, and professionals to take digital concepts from RTL simulation directly to physical silicon.

System **Demonstration**

1×1 "Compact"

160×120 logical grid | monochrome trails | cardinal movement

1×2 "Full"



640×480 framebuffer | color | 64 directions | boost and gaps

Two Independent Implementations

To evaluate area vs. features, we created two standalone ASIC projects.

Compact Achtung version (1x1)

Aggressively optimized to fit within one single Tiny Tapeout tile, achieving 92.6% utilization.

Uses a 1-bit monochrome occupancy map. The game advances on a 160x120 logical grid with orthogonal movement, expanded directly to 640x480 VGA output.

Full Achtung version (1x2)

Spans two tiles to support expanded gameplay mechanics and rich color presentation at 91.9% utilization.

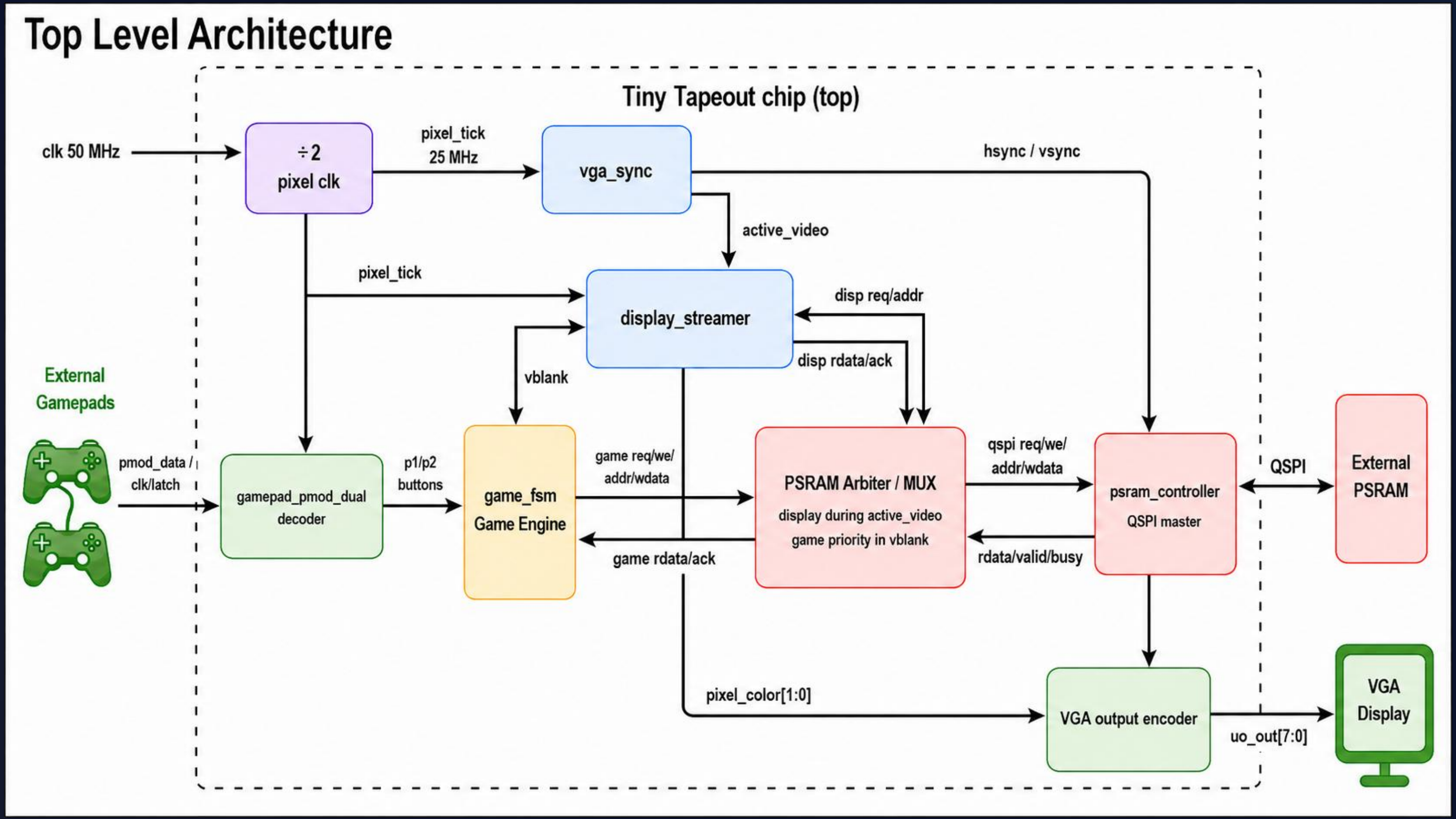
Operates directly on a full 640x480 framebuffer at 2 bits per pixel. Features 64-direction fixed-point movement, boost, and passable trail gaps.

Physical Hardware Components

To implement a full gaming system within ASIC boundaries, our design interfaces with several external components via the TT carrier board:

- **Tiny Tapeout ASIC:** The physical silicon hosting our synthesized Game FSM, VGA Sync, and memory arbitration logic.
- **External PSRAM (APS6404L): A high-speed, quad-SPI memory chip providing 8 MiB of external framebuffer memory.**
- **Gamepad PMOD:** A physical interface board allowing two standard SNES-style controllers to communicate serially with the ASIC.
- **VGA Breakout:** Drives the analog color signals and sync pulses directly to a standard 640x480 computer monitor.

Architecture Diagram



Top Level Architecture

Game FSM

Decodes asynchronous PMOD controller signals. Computes logical paths, evaluates collisions, and writes new trails to PSRAM strictly during vertical blanking.

VGA Sync

Generates 640x480 output timings. Enforces strict Time Division Multiplexing to safely arbitrate PSRAM access without visual screen tearing.

PSRAM Controller

Manages Quad SPI bursts to the APS6404L memory. Overlaps prefetching with pixel consumption to hide heavy communication protocol overhead.

Simplified Game State Machine

The functional game loop, with detailed RTL states grouped by purpose

1×1 skips the lobby; 1×2 adds colors, boost and gaps. READ is grouped into collision checking here.

Simplified functional view; the RTL retains separate LOBBY, CLEAR, IDLE, ANGLE, CHECK, READ, WRITE and OVER encodings.

The Memory Bottleneck

25 MHz

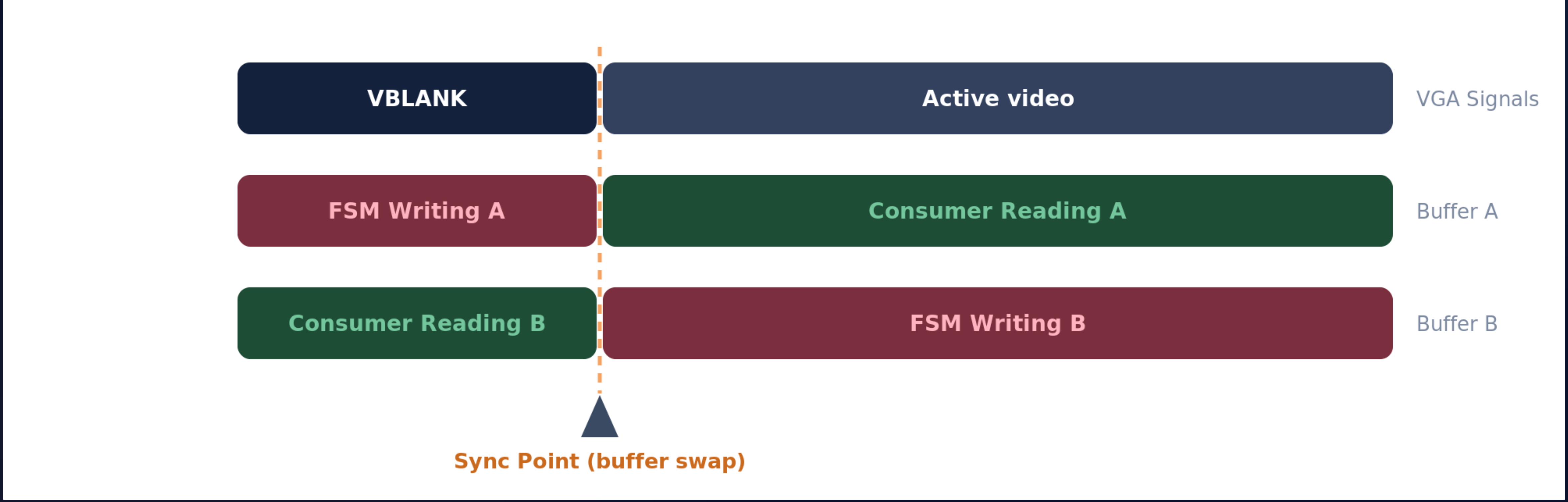
PSRAM SCLK LIMIT

Why direct reading fails:

While the PSRAM supports 133 MHz burst, Tiny Tapeout's output-switching limit sets the practical clock bound to roughly 25 MHz.

- A visible pixel is consumed on every 25 MHz cycle.
- PSRAM requires multiple cycles for Command, Address, and Wait phases before returning data.
- **The Fix:** We engineered packed framebuffer data combined with overlapped prefetching to ensure the display outpaces memory latency.

The Memory Bottleneck – Ping Pong Buffers



Buffer Sizing Mathematical Analysis

For the 1x2 full-game target (2 bits per physical pixel), we calculated the exact burst size (N bytes) required to prevent pixel starvation.

$$C_{disp} = 4N$$

$$C_{fetch} = 16 + 2N$$

$$\mathbf{C_{fetch} \le C_{disp} \implies 16 + 2N \le 4N \implies N \ge 8}$$

An 8-byte burst is the absolute minimum threshold to sustain a 640x480 VGA stream at 25 MHz.

Buffer Sizing & Resource Tradeoff

Choosing an 8-byte burst prevents display starvation while keeping flip-flop utilization safely within Tiny Tapeout's strict standard-cell constraints.

Buffer Size	Display Time (Cycles)	Fetch Time (Cycles)	Status	Naive Storage Cost
1 Byte (8-bit)	4	18	FAIL (Starvation)	16 FFs
4 Bytes (32-bit)	16	24	FAIL (Fetch lags)	64 FFs
8 Bytes (64-bit)	**32**	**32**	PASS (Chosen for 1x2)	**128 FFs (Optimal)**
16 Bytes (128-bit)	64	48	PASS (Margin)	256 FFs (Wasteful)

Verification Methodology

Verifying ASIC memory interactions requires rigorous, automated testing far beyond manual waveform inspection.

- **Python Behavioral Models:** Serve as an independent source of truth (test oracle).
- **Deterministic JSON Traces:** Record and execute repeatable, frame-by-frame gameplay inputs for stress testing.
- **Automated Equivalence Checking:** Compares final PSRAM framebuffer maps bit-for-bit between Python and RTL.
- **Subsystem Testing:** Includes pixel-exact VGA capture verification and full gate-level smoke tests.

Live Hardware Demonstration

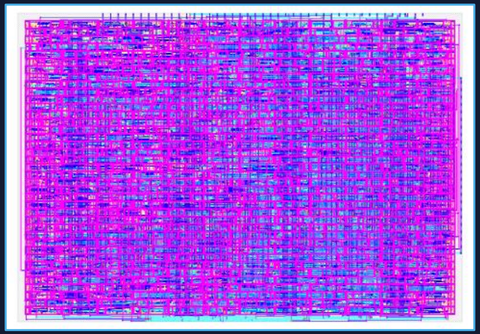
Synthesis & Layout Results

Both independent projects were successfully hardened using the open-source LibreLane flow with the sky130A PDK.

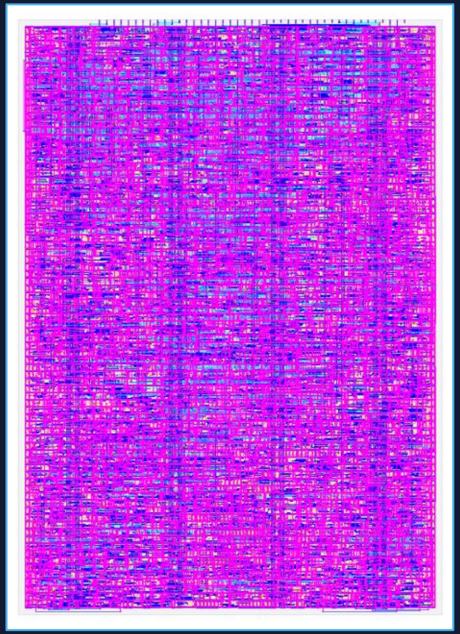
METRIC	1×1 "COMPACT" (TT_UM_JA_ACHTUNG_1X1)	1×2 "FULL" (TT_UM_JA_ACHTUNG_1X2)
Standard Cell Area	15,279.7 μm^2	31,468.9 μm^2
Die Utilization	92.64%	91.87%
Worst Setup Slack	+5.224 ns	+1.557 ns
Hard Violations (DRC/LVS)	0	0
TT Precheck & Gate-Level	15/15 Passed	15/15 Passed

Physical Layout (GDSII)

1×1 COMPACT



1×2 FULL



The final step before manufacturing is generating the physical geometric masks, known as the GDSII format.

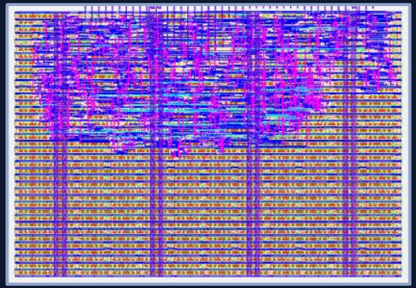
- **Standard Cell Placement:** Every logic gate from the RTL is physically placed within the designated 1x1 or 1x2 tile boundary.
- **Routing & Interconnects:** The OpenLane/LibreLane flow automatically routes microscopic metal layers to connect the components.
- **Density Optimization: Final utilization reached 92.64% (1×1) and 91.87% (1×2), leaving very little unused silicon.**
- **Signoff Checks: Confirms clean DRC geometry, LVS equivalence, antenna checks, and non-negative setup/hold timing before submission.**

Actual 2D GDS Density Comparison

Official TT renders beside the same purple/blue JA-Achtung GDS views used on the previous slide

1×1 TILE

PUBLIC TT REFERENCE

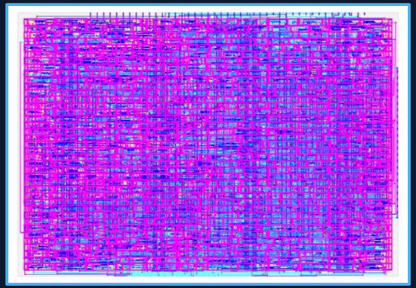


27.48%

Dummy Counter

Chinmay / TT08

JA-ACHTUNG GDS VIEW



92.64%

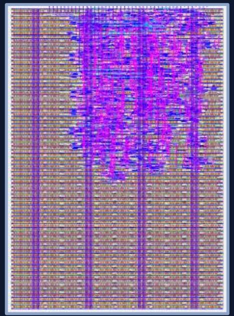
JA-Achtung Compact

final metric - layer-color view

+65.16 percentage points | ≈3.4× utilization

1×2 TILE

PUBLIC TT REFERENCE

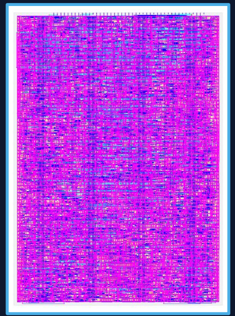


26.03%

UART

Darryl Miles / TT08

JA-ACHTUNG GDS VIEW



91.87%

JA-Achtung Full

final metric - layer-color view

+65.84 percentage points | ≈3.53× utilization

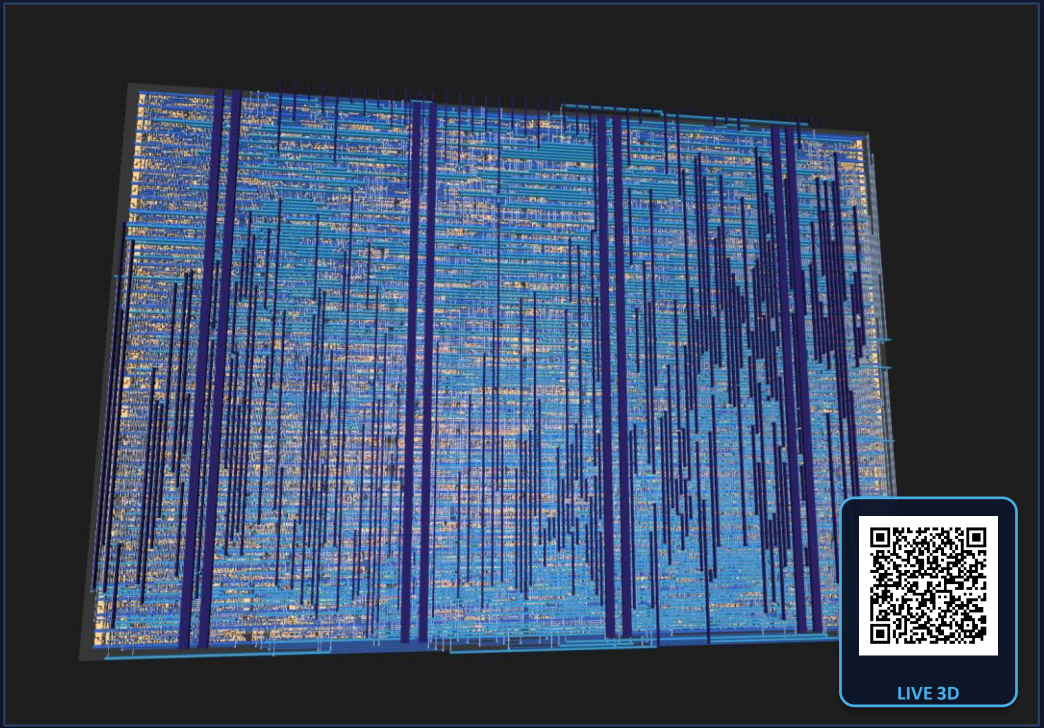
JA-Achtung fills almost the entire tile; the reference designs leave much more lightly used area.

References are official TT renders. JA-Achtung reuses the earlier layer-color view; percentages are final hardening metrics.

3D Physical Layout

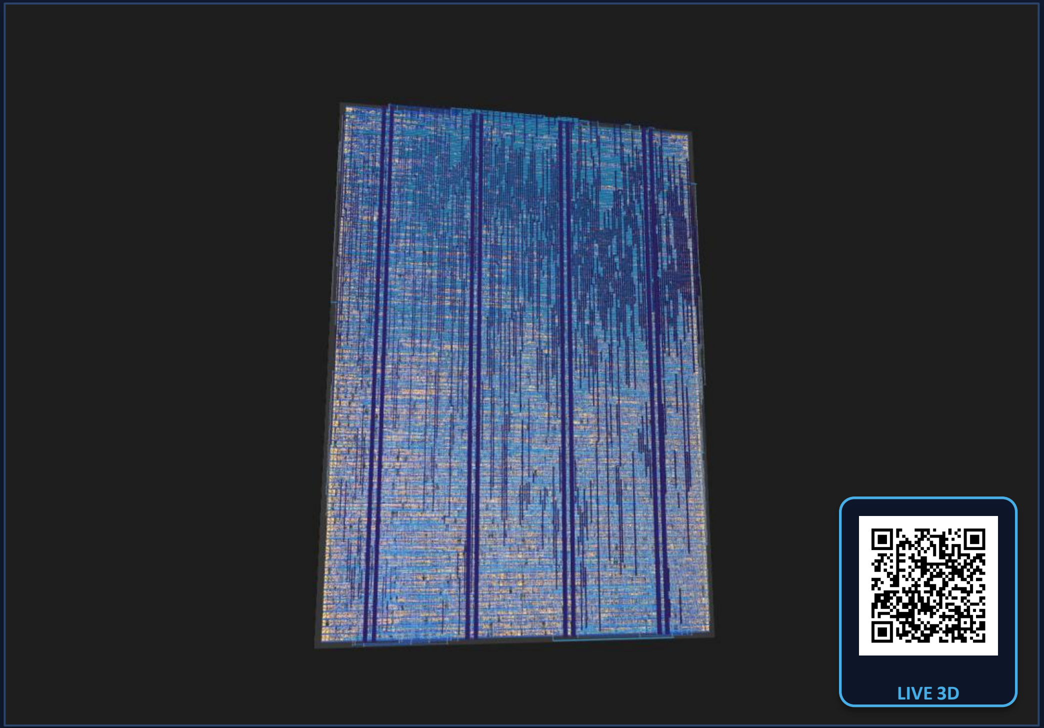
Interactive SKY130 layer-stack views generated from the final GDSII

1×1 COMPACT



FINAL GDS | SKY130

1×2 FULL



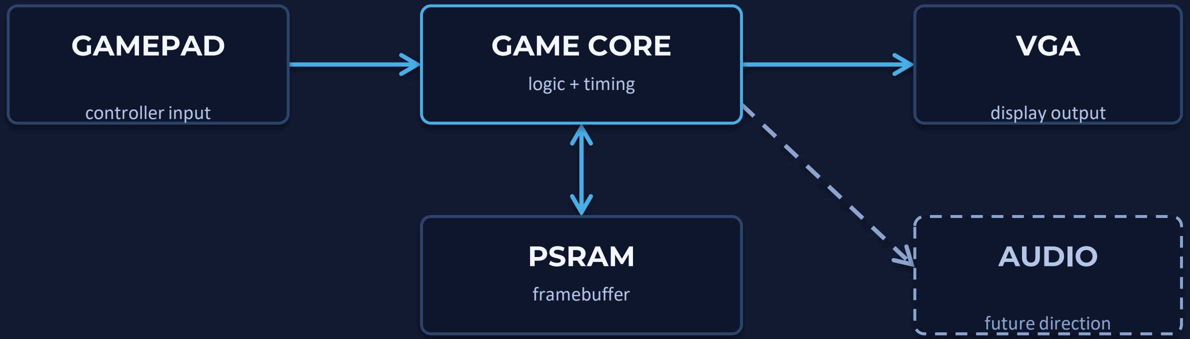
FINAL GDS | SKY130

Click or scan either render to open its live 3D viewer

Why This Project Stands Out

A distinctive combination of system integration, physical-design density, and evidence-driven optimization

ONE COMPACT SYSTEM



VGA output, controller input, and external PSRAM operate together as one real-time game pipeline.

A distinctive level of integration for a compact 1×1 / 1×2 Tiny Tapeout design.

HIGH-DENSITY SIGNOFF

92.64%	91.87%
1×1 COMPACT	1×2 FULL

0 hard DRC/LVS | 15/15 checks passed

AI + TT COMMUNITY

Discord guidance and AI analysis of hardening/GDS reports helped us:

- identify the real area drivers
- suggest targeted improvements
- compact both implementations

Every change was re-hardened and verified.

AI's strongest role: fast physical-design analysis - not a replacement for engineering judgment or verification.

How We Fit the Game into 1×1 / 1×2

We optimized synthesized hardware - not simply lines of source code.

MODULAR / GENERAL — COMPACT / ASIC-SPECIFIC

Purpose-built RTL reduced cells and timing depth; tests compensate for the lost abstraction.

DESIGN DECISION	AREA / TIMING WIN	TRADE-OFF
SELECTIVE ABSTRACTION	Share PSRAM control and buffers across module boundaries.	Tighter ownership assumptions
ALIGN + NARROW	Pack pixels/bursts; omit address bits known to be zero.	Fixed geometry and burst sizes
SHIFTS + SLICES	Shift-add and bit slices replace x20/x160 and /4,/8,/64.	Scaling assumptions enter RTL
ONE MUXED DATAPATH	Both players share movement, collision and address logic.	More manual selection logic
FIXED DIRECTION LUT	16 quarter-wave vectors generate all 64 headings.	Fixed precision; table upkeep

Smaller / faster hardware ↔ tighter coupling, fixed assumptions, less immediate readability — 92.64% | 91.87%

Comments, cocotb, deterministic replay, equivalence checks and repeated hardening kept each optimization safe.

Summary & Future Work

Preparing for physical silicon bring-up, validating I/O timing, and exploring analog enhancements.

Questions?

Thank you for your attention.

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